

Review Problem: Process Cost Flows and Costing Units

Luxguard Home Paint Company produces exterior latex paint, which it sells in one-gallon containers. The company has two processing departments—Base Fab and Finishing. White paint, which is used as a base for all the company's paints, is mixed from raw ingredients in the

Base Fab Department. Pigments are then added to the basic white paint, the pigmented paint is squirted under pressure into one-gallon containers, and the containers are labeled and packed for shipping in the Finishing Department. Information relating to the company's operations for April follows:

- Issued raw materials for use in production: Base Fab Department, \$851,000; and Finishing Department, \$629,000.
- Incurred direct labor costs: Base Fab Department, \$330,000; and Finishing Department, \$270,000.
- Applied manufacturing overhead cost: Base Fab Department, \$665,000; and Finishing Department, \$405,000.
- Transferred basic white paint from the Base Fab Department to the Finishing Department, \$1,850,000.
- Transferred paint that had been prepared for shipping from the Finishing Department to Finished Goods, \$3,200,000.

Required:

- Prepare journal entries to record items (a) through (e) above.
- Post the journal entries from (1) above to T-accounts. The balance in the Base Fab Department's Work in Process account on April 1 was \$150,000; the balance in the Finishing Department's Work in Process account was \$70,000. After posting entries to the T-accounts, find the ending balance in each department's Work in Process account.
- Determine the cost of ending work in process inventories and of units transferred out of the Base Fab Department in April. The following additional information is available regarding production in the Base Fab Department during April:

Production data:

Units (gallons) in process, April 1: materials 100% complete;	
labor and overhead 60% complete	30,000
Units (gallons) started into production during April	420,000
Units (gallons) completed and transferred to the	
Finishing Department	370,000
Units (gallons) in process, April 30: materials 50% complete;	
labor and overhead 25% complete	80,000

Cost data:

Work in process inventory, April 1:	
Materials	\$ 92,000
Labor	21,000
Overhead	37,000
Total cost of work in process inventory	<u>\$ 150,000</u>

Cost added during April:

Materials	\$ 851,000
Labor	330,000
Overhead	665,000
Total cost added during April	<u>\$1,846,000</u>

- Prepare a cost reconciliation report for April.

Solution to Review Problem

1.	a.	Work in Process—Base Fab Department	851,000	
		Work in Process—Finishing Department	629,000	
		Raw Materials		1,480,000
	b.	Work in Process—Base Fab Department	330,000	
		Work in Process—Finishing Department	270,000	
		Salaries and Wages Payable		600,000
	c.	Work in Process—Base Fab Department	665,000	
		Work in Process—Finishing Department	405,000	
		Manufacturing Overhead		1,070,000
	d.	Work in Process—Finishing Department	1,850,000	
		Work in Process—Base Fab Department		1,850,000
	e.	Finished Goods	3,200,000	
		Work in Process—Finishing Department		3,200,000

2.

Raw Materials

Bal.	XXX	(a)	1,480,000
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Work in Process— Base Fab Department

Bal.	150,000	(d)	1,850,000
(a)	851,000		
(b)	330,000		
(c)	665,000		
Bal.	146,000		

Work in Process—Finishing Department

Bal.	70,000	(e)	3,200,000
(a)	629,000		
(b)	270,000		
(c)	405,000		
(d)	1,850,000		
Bal.	24,000		

Salaries and Wages Payable

	(b)	600,000
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Manufacturing Overhead

(Various actual costs)	(c)	1,070,000
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Finished Goods

Bal.	XXX	
(e)	3,200,000	

3. First, we must compute the equivalent units of production for each cost category:

Base Fab Department Equivalent Units of Production			
	Materials	Labor	Overhead
Units transferred to the next department	370,000	370,000	370,000
Ending work in process inventory (materials: 80,000 units × 50% complete; labor: 80,000 units × 25% complete; overhead: 80,000 units × 25% complete)	40,000	20,000	20,000
Equivalent units of production	<u>410,000</u>	<u>390,000</u>	<u>390,000</u>

Then we must compute the cost per equivalent unit for each cost category:

Base Fab Department Costs per Equivalent Unit			
	Materials	Labor	Overhead
Costs:			
Cost of beginning work in process inventory	\$ 92,000	\$ 21,000	\$ 37,000
Costs added during the period	<u>851,000</u>	<u>330,000</u>	<u>665,000</u>
Total cost (a)	<u>\$943,000</u>	<u>\$351,000</u>	<u>\$702,000</u>
Equivalent units of production (b)	410,000	390,000	390,000
Cost per equivalent unit (a) ÷ (b)	\$2.30	\$0.90	\$1.80

The costs per equivalent unit can then be applied to the units in ending work in process inventory and the units transferred out as follows:

Base Fab Department Costs of Ending Work in Process Inventory and the Units Transferred Out				
	Materials	Labor	Overhead	Total
Ending work in process inventory:				
Equivalent units of production	40,000	20,000	20,000	
Cost per equivalent unit	\$2.30	\$0.90	\$1.80	
Cost of ending work in process inventory	\$92,000	\$18,000	\$36,000	\$146,000
Units completed and transferred out:				
Units transferred to the next department	370,000	370,000	370,000	
Cost per equivalent unit	\$2.30	\$0.90	\$1.80	
Cost of units completed and transferred out	\$851,000	\$333,000	\$666,000	\$1,850,000

4.

Base Fab Department Cost Reconciliation	
Costs to be accounted for:	
Cost of beginning work in process inventory	\$ 150,000
Costs added to production during the period	<u>1,846,000</u>
Total cost to be accounted for	<u>\$1,996,000</u>
Costs accounted for as follows:	
Cost of ending work in process inventory	\$ 146,000
Cost of units transferred out	<u>1,850,000</u>
Total cost accounted for	<u>\$1,996,000</u>